

```
from google.colab import drive
```

```
drive.mount("/content/drive")
```

↗ Mounted at /content/drive

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

```
from sklearn.cluster import KMeans, k_means
from sklearn.decomposition import PCA
```

```
df=pd.read_csv("/content/drive/MyDrive/sales_data_sample.csv", sep="," , encoding="latin-1")
```

```
df.head()
```

↗

	ORDERNUMBER	QUANTITYORDERED	PRICEEACH	ORDERLINENUMBER	SALES	ORDERDATE	STATUS	QTR_ID	
0	10107	30	95.70	2	2871.00	2/24/2003 0:00	Shipped	1	
1	10121	34	81.35	5	2765.90	5/7/2003 0:00	Shipped	2	
2	10134	41	94.74	2	3884.34	7/1/2003 0:00	Shipped	3	
3	10145	45	83.26	6	3746.70	8/25/2003 0:00	Shipped	3	
4	10159	49	100.00	14	5205.27	10/10/2003 0:00	Shipped	4	

5 rows × 25 columns

```
df.shape
```

↗ (2823, 25)

```
df.describe()
```



	ORDERNUMBER	QUANTITYORDERED	PRICEEACH	ORDERLINENUMBER	SALES	QTR_ID	MONTH_ID
count	2823.000000	2823.000000	2823.000000	2823.000000	2823.000000	2823.000000	2823.000000
mean	10258.725115	35.092809	83.658544	6.466171	3553.889072	2.717676	7.092105
std	92.085478	9.741443	20.174277	4.225841	1841.865106	1.203878	3.651181
min	10100.000000	6.000000	26.880000	1.000000	482.130000	1.000000	1.000000
25%	10180.000000	27.000000	68.860000	3.000000	2203.430000	2.000000	4.000000
50%	10262.000000	35.000000	95.700000	6.000000	3184.800000	3.000000	8.000000
75%	10333.500000	43.000000	100.000000	9.000000	4508.000000	4.000000	11.000000
max	10425.000000	97.000000	100.000000	18.000000	14082.800000	4.000000	12.000000

```
df.info()
```



```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2823 entries, 0 to 2822
Data columns (total 25 columns):
#   Column                Non-Null Count  Dtype
---  -
0   ORDERNUMBER            2823 non-null   int64
1   QUANTITYORDERED        2823 non-null   int64
2   PRICEEACH              2823 non-null   float64
3   ORDERLINENUMBER        2823 non-null   int64
4   SALES                   2823 non-null   float64
5   ORDERDATE              2823 non-null   object
6   STATUS                 2823 non-null   object
7   QTR_ID                 2823 non-null   int64
8   MONTH_ID              2823 non-null   int64
9   YEAR_ID               2823 non-null   int64
10  PRODUCTLINE            2823 non-null   object
11  MSRP                   2823 non-null   int64
12  PRODUCTCODE            2823 non-null   object
13  CUSTOMERNAME           2823 non-null   object
14  PHONE                  2823 non-null   object
15  ADDRESSLINE1           2823 non-null   object
16  ADDRESSLINE2           302 non-null    object
17  CITY                   2823 non-null   object
18  STATE                  1337 non-null   object
19  POSTALCODE             2747 non-null   object
20  COUNTRY                2823 non-null   object
21  TERRITORY              1749 non-null   object
22  CONTACTLASTNAME        2823 non-null   object
23  CONTACTFIRSTNAME       2823 non-null   object
24  DEALSIZE               2823 non-null   object
dtypes: float64(2), int64(7), object(16)
memory usage: 551.5+ KB
```

```
df.isnull().sum()
```



0

ORDERNUMBER	0
QUANTITYORDERED	0
PRICEEACH	0
ORDERLINENUMBER	0
SALES	0
ORDERDATE	0
STATUS	0
QTR_ID	0
MONTH_ID	0
YEAR_ID	0
PRODUCTLINE	0
MSRP	0
PRODUCTCODE	0
CUSTOMERNAME	0
PHONE	0
ADDRESSLINE1	0
ADDRESSLINE2	2521
CITY	0
STATE	1486
POSTALCODE	76
COUNTRY	0
TERRITORY	1074
CONTACTLASTNAME	0
CONTACTFIRSTNAME	0
DEALSIZE	0

dtype: int64

df.dtypes



0

ORDERNUMBER	int64
QUANTITYORDERED	int64
PRICEEACH	float64
ORDERLINENUMBER	int64
SALES	float64
ORDERDATE	object
STATUS	object
QTR_ID	int64
MONTH_ID	int64
YEAR_ID	int64
PRODUCTLINE	object
MSRP	int64
PRODUCTCODE	object
CUSTOMERNAME	object
PHONE	object
ADDRESSLINE1	object
ADDRESSLINE2	object
CITY	object
STATE	object
POSTALCODE	object
COUNTRY	object
TERRITORY	object
CONTACTLASTNAME	object
CONTACTFIRSTNAME	object
DEALSIZE	object

dtype: object

```
df_drop = ['ADDRESSLINE1', 'ADDRESSLINE2', 'STATUS', 'POSTALCODE', 'CITY', 'TERRITORY', 'PHONE', '']  
df = df.drop(df_drop, axis=1)
```

```
df.isnull().sum()
```



	0
QUANTITYORDERED	0
PRICEEACH	0
ORDERLINENUMBER	0
SALES	0
ORDERDATE	0
QTR_ID	0
MONTH_ID	0
YEAR_ID	0
PRODUCTLINE	0
MSRP	0
PRODUCTCODE	0
COUNTRY	0
DEALSIZE	0

dtype: int64

df.dtypes



	0
QUANTITYORDERED	int64
PRICEEACH	float64
ORDERLINENUMBER	int64
SALES	float64
ORDERDATE	object
QTR_ID	int64
MONTH_ID	int64
YEAR_ID	int64
PRODUCTLINE	object
MSRP	int64
PRODUCTCODE	object
COUNTRY	object
DEALSIZE	object

dtype: object

df['COUNTRY'].unique()



```
array(['USA', 'France', 'Norway', 'Australia', 'Finland', 'Austria', 'UK',  
      'Spain', 'Sweden', 'Singapore', 'Canada', 'Japan', 'Italy',
```

```
'Denmark', 'Belgium', 'Philippines', 'Germany', 'Switzerland',  
'Ireland'], dtype=object)
```

```
df['PRODUCTLINE'].unique()
```

```
➦ array(['Motorcycles', 'Classic Cars', 'Trucks and Buses', 'Vintage Cars',  
        'Planes', 'Ships', 'Trains'], dtype=object)
```

```
df['DEALSIZE'].unique()
```

```
➦ array(['Small', 'Medium', 'Large'], dtype=object)
```

```
productline = pd.get_dummies(df['PRODUCTLINE'])  
Dealsize = pd.get_dummies(df['DEALSIZE'])
```

```
df = pd.concat([df,productline,Dealsize], axis = 1)
```

```
df_drop = ['COUNTRY','PRODUCTLINE','DEALSIZE']  
df = df.drop(df_drop, axis=1)
```

```
df['PRODUCTCODE'] = pd.Categorical(df['PRODUCTCODE']).codes
```

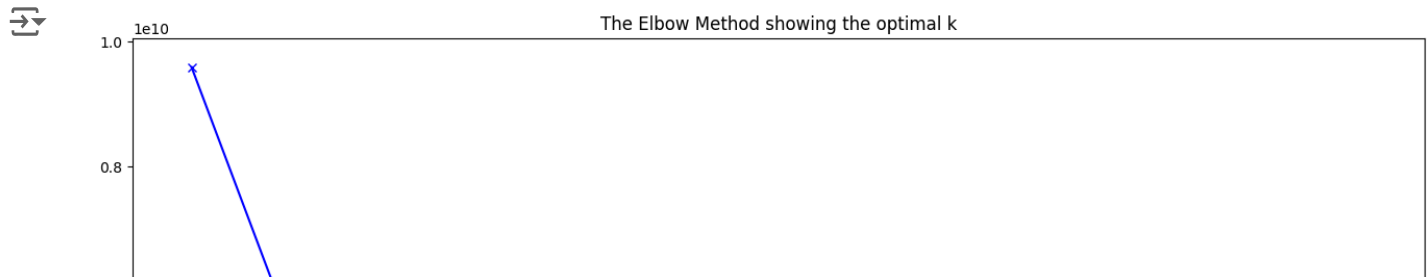
```
df.drop('ORDERDATE', axis=1, inplace=True)
```

```
df.dtypes
```

**dtype:** object[illegible]

```
super()._check_params_vs_input(X, default_n_init=10)
```

```
plt.figure(figsize=(16,8))
plt.plot(K, distortions, 'bx-')
plt.xlabel('k')
plt.ylabel('Distortion')
plt.title('The Elbow Method showing the optimal k')
plt.show()
```



```
X_train = df.values
```

```
X_train.shape
```

```
(2823, 19)
```

```
model = KMeans(n_clusters=3,random_state=2)
model = model.fit(X_train)
predictions = model.predict(X_train)
```

```
/usr/local/lib/python3.10/dist-packages/sklearn/cluster/_kmeans.py:1416: FutureWarning: The de-
super()._check_params_vs_input(X, default_n_init=10)
```

```
unique,counts = np.unique(predictions,return_counts=True)
```

```
counts = counts.reshape(1,3)
```

```
counts_df = pd.DataFrame(counts,columns=['Cluster1','Cluster2','Cluster3'])
```

```
counts_df.head()
```

```
Cluster1 Cluster2 Cluster3
0      1360      375     1088
```

```
pca = PCA(n_components=2)
```

```
reduced_X = pd.DataFrame(pca.fit_transform(X_train),columns=['PCA1','PCA2'])
```

```
reduced_X.head()
```

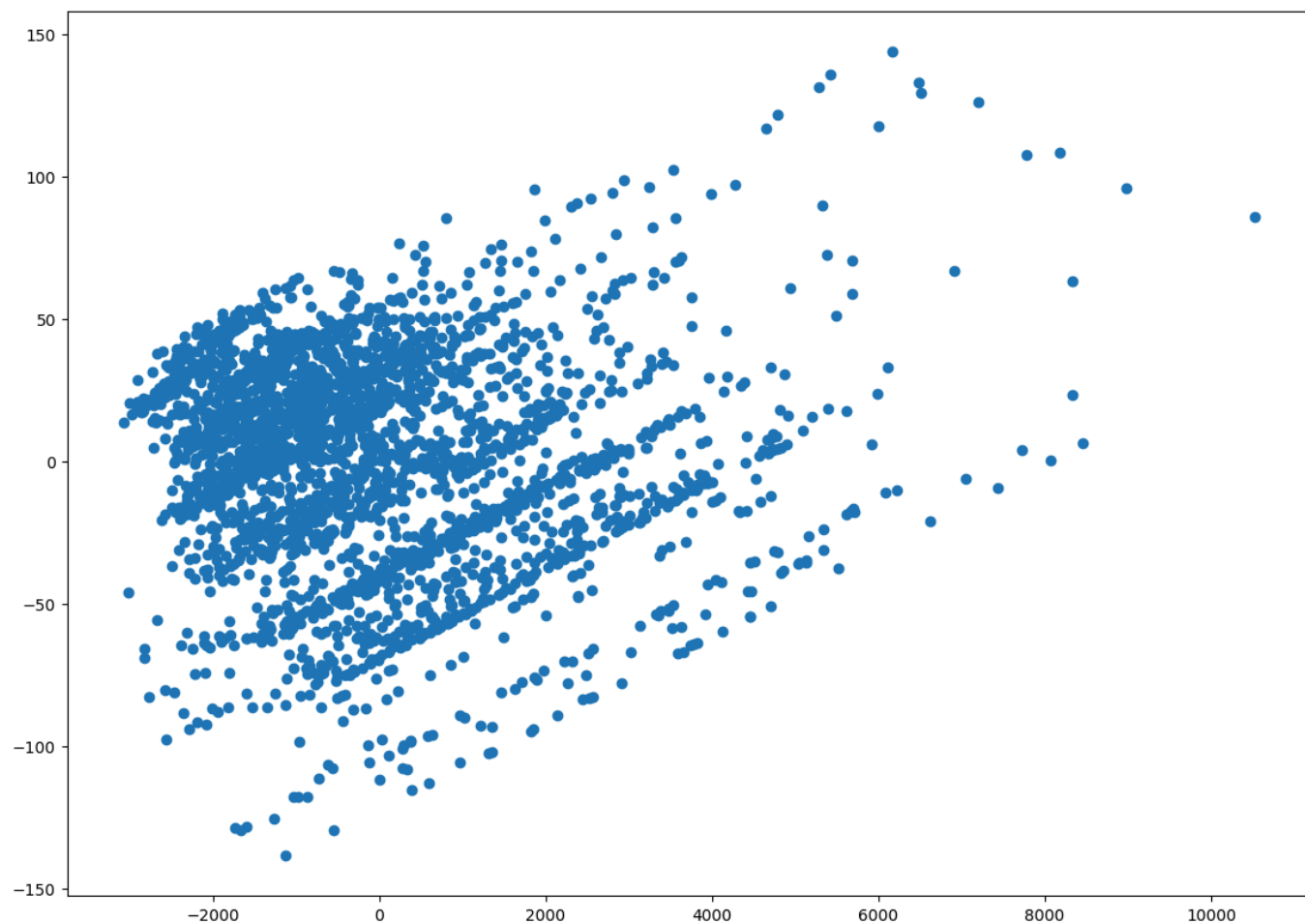



	PCA1	PCA2
0	-682.488323	-42.819535
1	-787.665502	-41.694991
2	330.732170	-26.481208
3	193.040232	-26.285766
4	1651.532874	-6.891196

```
plt.figure(figsize=(14,10))
plt.scatter(reduced_X['PCA1'],reduced_X['PCA2'])
```



<matplotlib.collections.PathCollection at 0x7cdd943982b0>



model.cluster_centers_



```
array([[ 3.07860294e+01,  7.00256985e+01,  6.66985294e+00,
        2.11891932e+03,  2.71764706e+00,  7.09338235e+00,
        2.00381397e+03,  7.84757353e+01,  6.25088235e+01,
        2.64705882e-01,  1.21323529e-01,  1.29411765e-01,
        1.00735294e-01,  3.75000000e-02,  9.26470588e-02,
        2.53676471e-01, -1.06858966e-15,  5.73529412e-02,
        9.42647059e-01],
       [ 4.46106667e+01,  9.98936800e+01,  5.78400000e+00,
        7.08806629e+03,  2.70133333e+00,  7.03733333e+00,
```

```

2.00389067e+03, 1.45581333e+02, 3.17040000e+01,
5.33333333e-01, 1.06666667e-01, 7.20000000e-02,
2.13333333e-02, 1.33333333e-02, 1.30666667e-01,
1.22666667e-01, 4.18666667e-01, 5.81333333e-01,
-2.72004641e-15],
[ 3.71957721e+01, 9.51038511e+01, 6.44669118e+00,
4.12947951e+03, 2.72334559e+00, 7.11029412e+00,
2.00379044e+03, 1.13051471e+02, 5.04604779e+01,
3.74080882e-01, 1.15808824e-01, 9.46691176e-02,
8.18014706e-02, 1.93014706e-02, 1.15808824e-01,
1.98529412e-01, 2.08166817e-17, 1.00000000e+00,
-4.16333634e-15]])

```

```
reduced_centers = pca.transform(model.cluster_centers_)
```

```
reduced_centers
```

```

array([[ -1.43523424e+03,  2.54268709e+00],
       [  3.53456611e+03,  3.35030029e+00],
       [  5.75787016e+02, -4.33310390e+00]])

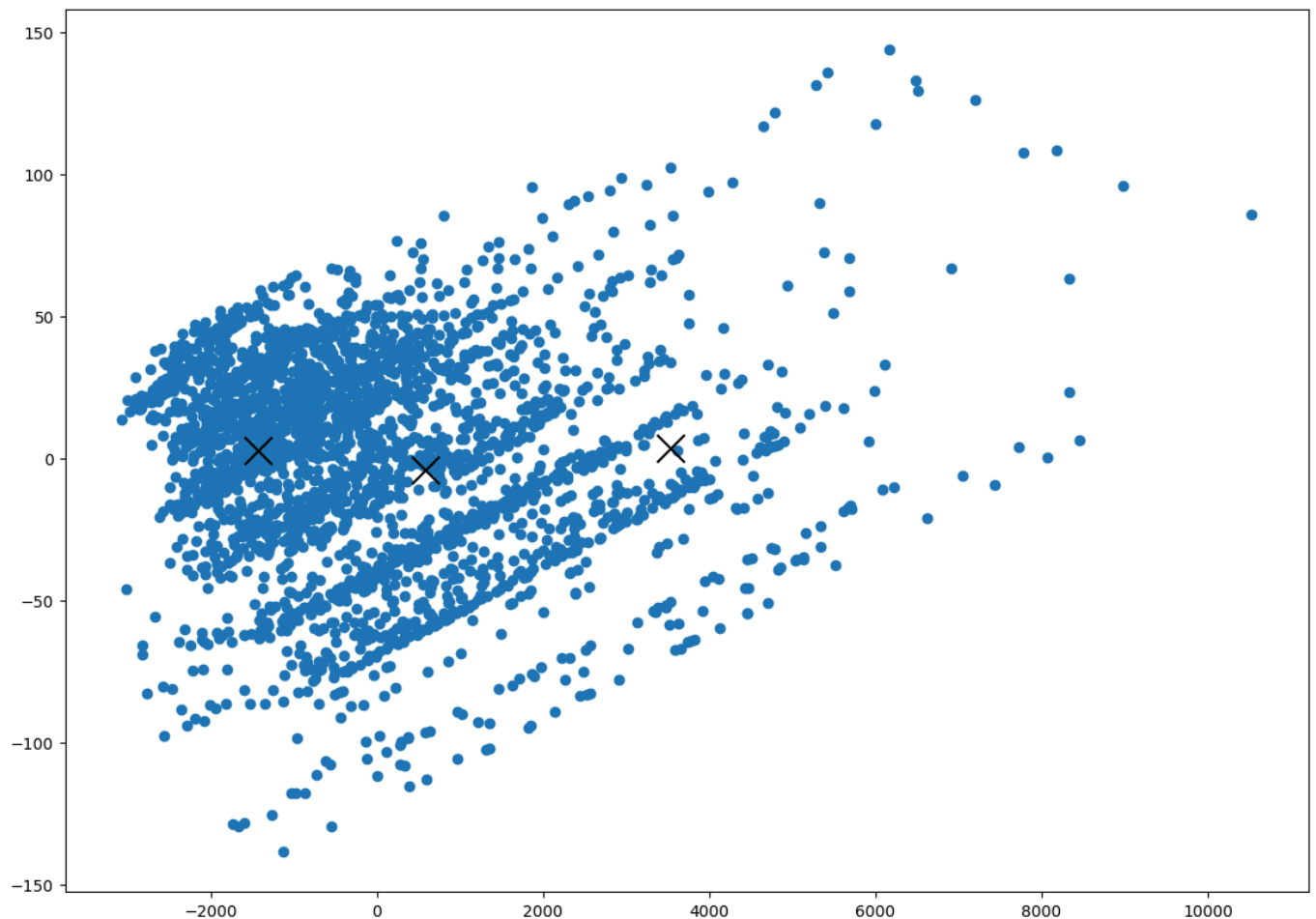
```

```

plt.figure(figsize=(14,10))
plt.scatter(reduced_X['PCA1'],reduced_X['PCA2'])
plt.scatter(reduced_centers[:,0],reduced_centers[:,1],color='black',marker='x',s=300)

```

```
<matplotlib.collections.PathCollection at 0x7cdd94f23cd0>
```



```
reduced_X['Clusters'] = predictions
```

```
reduced_X.head()
```